



MATHEMATICS

2012 Preliminary Course Final Examination

September 5, 2012

General instructions

- Working time – $2\frac{1}{2}$ hours.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 9)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NAME:

BOOKLETS USED:

Class (please ✓)

☐ 12M2A – Mr Lowe

Marker's use only.

QUESTION	1-8	9	10	11	12	13	Total	%
MARKS	$\frac{8}{8}$	$\frac{16}{16}$	$\frac{14}{14}$	$\frac{14}{14}$	$\frac{17}{17}$	$\frac{14}{14}$	$\frac{83}{83}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. What is the value of the sum of exterior angles of a polygon? **1**

(A) 180° (C) 90°
(B) 360° (D) None of these

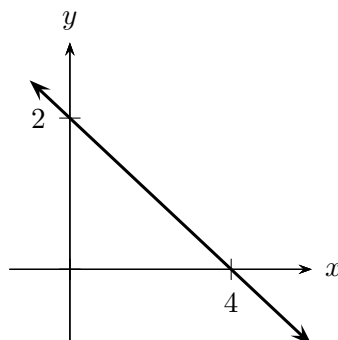
2. Which of the following is *not* a condition for congruent triangles? **1**

(A) SSS (C) SAS
(B) AAA (D) AAS

3. Which of these conditions for the gradient make two lines parallel? **1**

(A) Gradients equal. (C) Gradients are reciprocals
(B) Gradients multiply to -1 (D) None of these

4. Which of the following is the correct equation for the line shown in the diagram? **1**



(A) $y = -2x + 2$ (C) $y = 2x + 2$
(B) $y = -\frac{1}{2}x + 2$ (D) $y = \frac{1}{2}x + 2$

5. Which of the following makes the statement true? **1**

The sum of the interior angles of an n sided polygon is straight angles.

(A) $2n$ (C) $n - 1$
(B) n (D) $n - 2$

6. Which of the following makes the statement true? 1

Each angle of triangle is 60° .

- (A) an equilateral (C) an isosceles
(B) a scalene (D) an acute

7. What is the value of 1.9926 to two significant figures? 1

- (A) 2.0 (C) 2.09
(B) 1.9 (D) 2.01

8. What is the value of $|-8| - |10|$? 1

- (A) 2 (C) -1
(B) 1 (D) -2

End of Section I.
Examination continues overleaf.

Section II: Short answer

Question 9 (16 Marks) Commence a NEW page. **Marks**

- (a) Simplify $\sqrt{12} + \sqrt{3} - \sqrt{18}$. **2**
- (b) Express $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$ in simplest form with a rational denominator. **2**
- (c) Solve for x :
- i. $9^x = 243$ **2**
 - ii. $x = \log_8 \frac{1}{64}$ **2**
 - iii. $\frac{1-x}{3} + \frac{x-1}{2} = 1$ **3**
- (d) Solve $|3x - 2| \leq 4$. **2**
- (e) Solve for x and y simultaneously: **3**

$$\begin{cases} x + 2y = 1 \\ 3x + y = 13 \end{cases}$$

Question 10 (14 Marks) Commence a NEW page. **Marks**

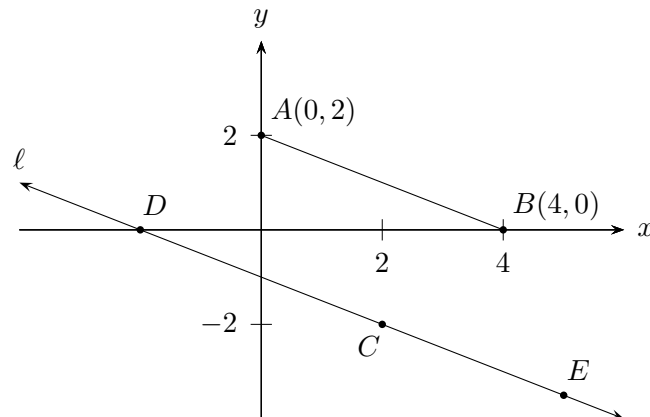
- (a) Write $\frac{a^2 - b^2}{a^{-1} + b^{-1}}$ without negative indices. **3**
- (b) Simplify:
- i. $1 - \frac{m}{m-n}$ **2**
 - ii. $2^n + 2^{n+1}$ **2**
 - iii. $\log_{12} \frac{2}{7} + \log_{12} \frac{7}{2}$ **2**
- (c) For the function
- $$f(x) = \begin{cases} 1-x & -2 \leq x \leq 0 \\ x^2 + 2 & 0 < x \leq 2 \end{cases}$$
- i. Draw a neat sketch of $f(x)$. **3**
 - ii. Write down the domain and range $f(x)$. **2**

Question 11 (14 Marks)

Commence a NEW page.

Marks

- (a) Refer to the following diagram:



- i. Find the gradient of AB . **1**
 - ii. Show that $AB = 2\sqrt{5}$ units. **1**
 - iii. Show that the equation of the line ℓ , through C parallel to AB is **1**

$$x + 2y + 2 = 0$$
 - iv. Find the coordinates of the point D where ℓ cuts the x axis. **1**
 - v. Show that the perpendicular distance from A to the ℓ is $\frac{6}{\sqrt{5}}$ units. **1**
 - vi. If the area of the trapezium $ABED$ is 18 units^2 , find the length of DE . **3**
- (b) Find the equation of the line passing through the intersection of $2x + 3y + 5 = 0$ and $3x - 4y - 3 = 0$, and through the point $(1, 2)$. **3**
- (c) Sketch the region defined by **3**

$$y > 2 - x \quad \text{and} \quad y \leq x^2$$

Question 12 (18 Marks)

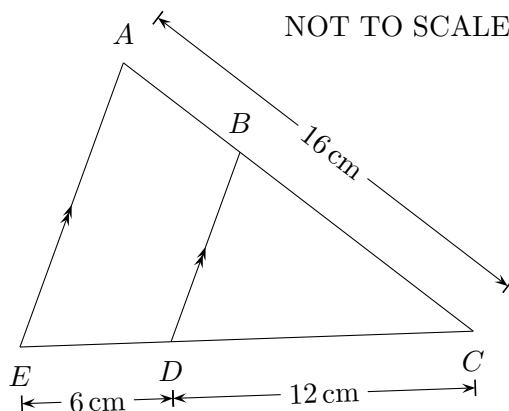
Commence a NEW page.

Marks

- (a) A quadrilateral $ABCD$ is such that $\triangle ABD$ and $\triangle BDC$ are equilateral.

- i. Draw a neat diagram to represent this information. **1**
- ii. Prove that $\angle BAC = 30^\circ$. **2**

- (b) The diagram shows $\triangle ACE$ with $AE \parallel BD$, $AC = 16$ cm, $CD = 12$ cm and $DE = 6$ cm.



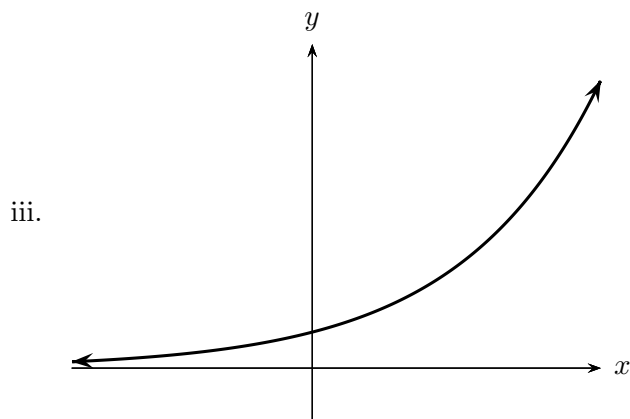
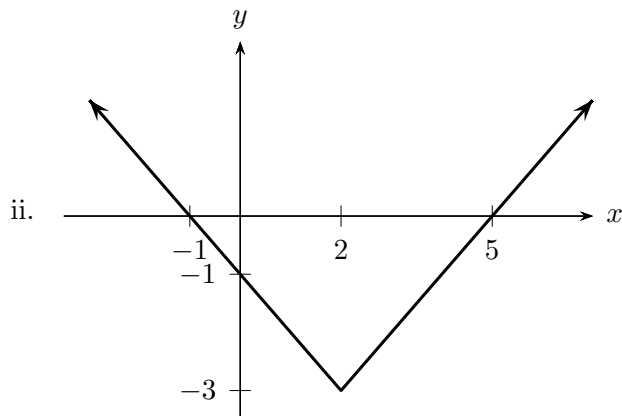
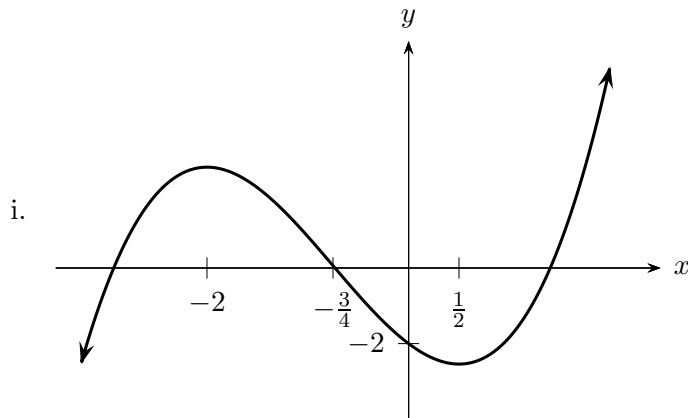
- i. Prove that $\triangle ACE$ is similar to $\triangle BCD$. **3**
 - ii. Hence or otherwise, find the length of AB . **2**
- (c) A rhombus with diagonals x and $x + 2$ cm has an area of 24 cm^2 . **3**

Find the value of x , and hence the value of the other diagonal.

Question 12 continued on the next page ...

Question 12 continued from page 6...

- (d) Copy the following diagrams into your writing booklet, and on the same set of axes, sketch the gradient function of the following:

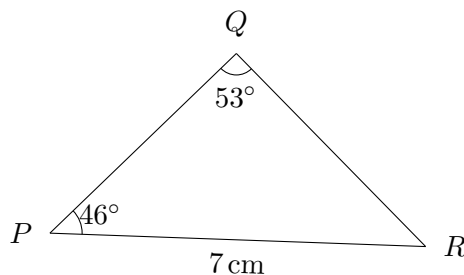


Question 13 (14 Marks)

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Marks

- (a) In $\triangle PQR$, $\angle QPR = 46^\circ$, $\angle PQR = 53^\circ$ and $PR = 7$ cm.



- | | | |
|-----|--------------------------------|---|
| i. | Find the length of QR . | 2 |
| ii. | Find the area of the triangle. | 2 |
- (b) Solve the following equation for θ , where $0^\circ \leq \theta \leq 360^\circ$:
- | | | |
|------|------------------------------------|---|
| i. | $\cos \theta = \frac{1}{\sqrt{2}}$ | 2 |
| ii. | $\tan \theta + 1 = 0$. | 2 |
| iii. | $\sin^2 x = \frac{3}{4}$ | 2 |
- (c) Consider the function $y = 3 \sin 2x$, where $0^\circ \leq x \leq 360^\circ$.
- | | | |
|-----|---|---|
| i. | State the period and amplitude of the function. | 2 |
| ii. | Draw a neat sketch of the function. | 2 |

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NAME:

Class (please ✓)

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- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |

12th Yearly Suggested Solns

- 1) B
- 2) B
- 3) ~~B~~ A
- 4) B
- 5) ~~D~~
- 6) A
- 7) A
- 8) D

9) a) $2\sqrt{3} + \sqrt{3} - 3\sqrt{2}$ 2
 $= 3\sqrt{3} - 3\sqrt{2}$

b) $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$ 2
 $= (\sqrt{3} + \sqrt{2})^2$

c) i) $3^{2x} = 243$
 $3^{2x} = 3^5$
 $2x = 5$ 2
 $x = \frac{5}{2}$

ii) $x = \log_8 8^{-2}$ 2
 $= -2$

iii) $2 - 2x + 3x - 3 = 6$
 $x - 1 = 6$ 3
 $x = 7$

d) $-4 \leq 3x - 2 \leq 4$
 $-2 \leq 3x \leq 6$
 $-\frac{2}{3} \leq x \leq 2$ 2

e) $x + 2y = 1$
 $6x + 2y = 26$

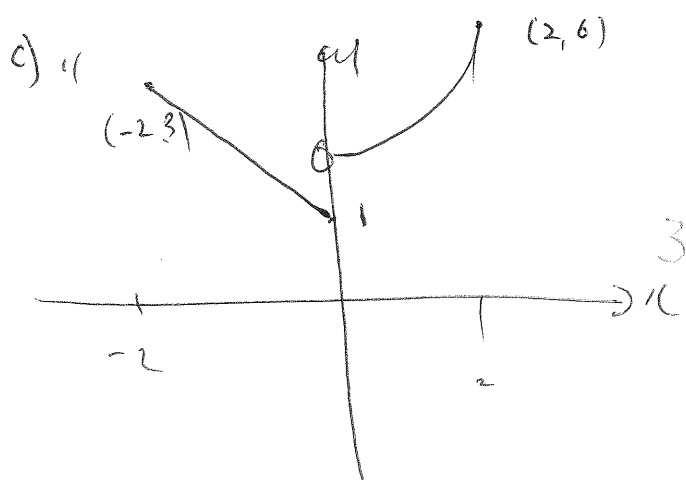
 $-5x = -25$
 $x = 5$
 $y = -2$ 3
 $(x, y) = (5, -2)$

10) a) $(a-b)(a+b)$ 1
 $\frac{\frac{1}{a} + \frac{1}{b}}{(a-b)(a+b)}$ 1
 $\frac{ab}{ab(a-b)}$ 1
 $= \frac{ab}{ab(a-b)}$

b) i) $\frac{m-n-m}{m-n}$ 2
 $= \frac{-n}{m-n} = \frac{n}{n-m}$

ii) $2^n + 2 \times 2^n$ 1
 $= 2^n (3)$ 1
 $= 3 \times 2^n$

iii) $\log_{12} 1$ 1
 $= 0$



ii) Domain $-2 \leq x \leq 2$

Range $1 \leq y \leq 6$

(ii) a) i) $m_{AB} = -\frac{1}{2}$

ii) $AB = \sqrt{2^2 + 4^2}$

$= \sqrt{20}$
 $= 2\sqrt{5}$

iii) $y - (-1) = -\frac{1}{2}(x - 4)$

$2y + 4 = -x + 4$

$x + 2y + 4 = 0$

iv) $y = 0$

$x = -2$

D is $(-2, 0)$

v) $d = \frac{|0 + 2(-2) + 4|}{\sqrt{1^2 + 2^2}}$

$= \frac{0}{\sqrt{5}}$

vi) Area $= \frac{1}{2}(AB + DE)$

$18 = \frac{3}{\sqrt{5}}(2\sqrt{5} + DE)$

$18 = 6 + \frac{3}{\sqrt{5}} DE$

$12 = \frac{3}{\sqrt{5}} DE$

$DE = 4\sqrt{5}$

b) line $\rightarrow$

$2x + 3y + 5 + k(3x - 4y - 3) = 0$

Through $(1, 2)$

$2 + 6 + 5 + k(3 - 8 - 3) = 0$

$13 - 8k = 0$

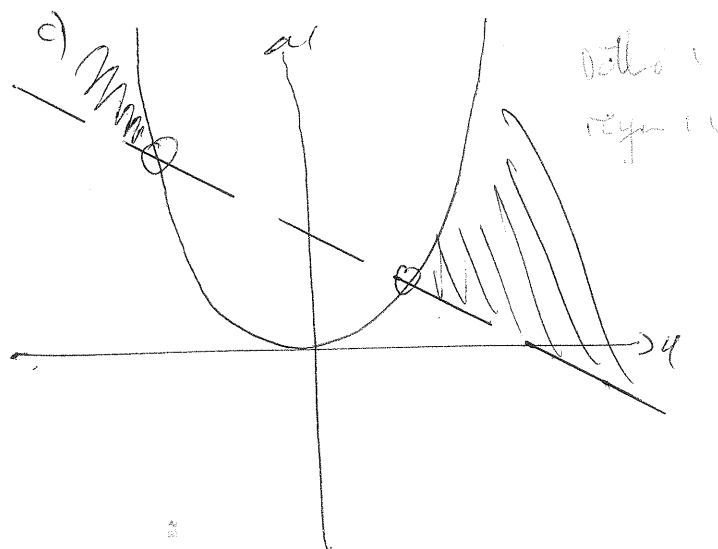
$8k = 13$

$k = \frac{13}{8}$

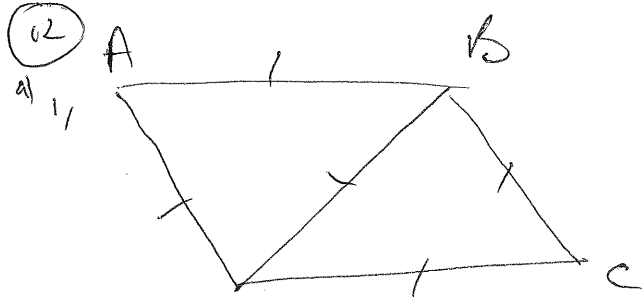
$\therefore 2x + 3y + 5 + \frac{13}{8}(3x - 4y - 3) = 0$

$16x + 24y + 40 + 39x - 52y - 39 = 0$

$55x - 28y + 1 = 0$



Q2



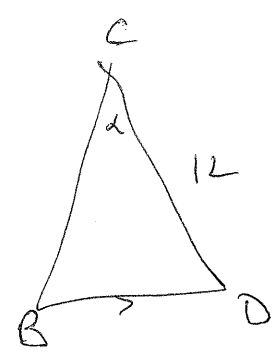
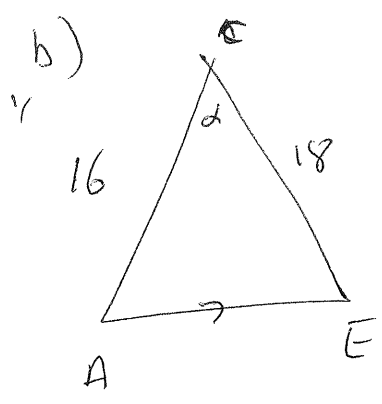
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$\angle BAD = 60^\circ$ [data]

ABCD is a rhombus as all sides equal

$\therefore AC$ bisects $\angle BAD$

$$\angle BAC = 30^\circ$$



In $\triangle ACE$ & $\triangle BCD$

$\angle C$ is common

$\angle CAB = \angle CBD$ (corresponding $\parallel$ lines)

$\therefore \triangle ACE \parallel \triangle BCD$ (AAA)

11/

$$\frac{BC}{16} = \frac{x}{18} \cdot \frac{2}{3}$$

$$BC = \frac{32}{3}$$

$$\therefore AB = 16 - \frac{32}{3}$$

$$= \frac{16}{3}$$

c)

$$\frac{1}{2} x(x+2) = 24$$

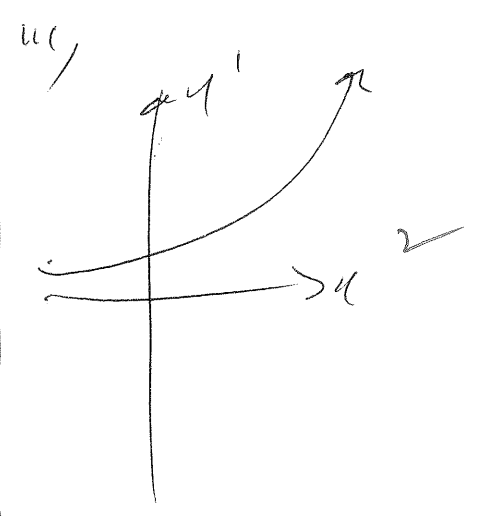
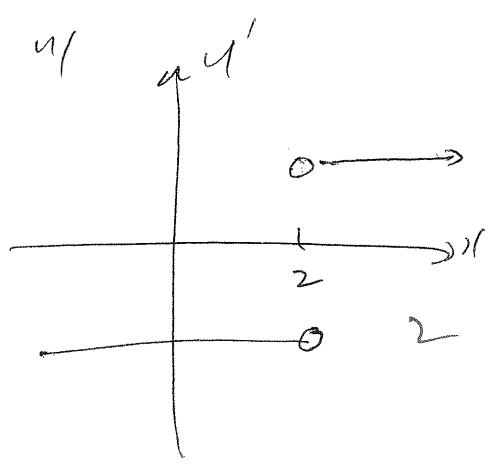
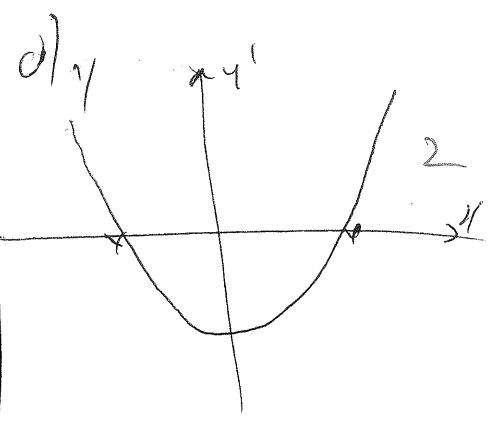
$$x^2 + 2x - 48 = 0$$

$$(x+8)(x-6) = 0$$

$$x = 6 \text{ or } -8$$

$$x > 0$$

$\therefore$ Diagonals are $6 \text{ cm} + 8 \text{ cm}$



(13) a) $\frac{QR}{\sin 46} = \frac{7}{\sin 53}$

$$QR = \frac{7 \sin 46}{\sin 53}$$

$$= \underline{6.3 \text{ cm}}$$

ii) $A = \frac{1}{2} \times QR \times RP \times \sin 81$

$$= \underline{21.8 \text{ cm}^2}$$

b) i) $\cos \theta = \frac{1}{\sqrt{2}}$ $\frac{s}{r} \left| \frac{A}{c} \right|$

Acute $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$

$\therefore \theta = 45^\circ, 315^\circ$ 2

ii) $\tan \theta = -1$ $\frac{\sqrt{s}}{r} \left| \frac{A}{c} \right|$

Acute $\tan^{-1}(1) = 45$

$\theta = 135^\circ, 315^\circ$ 2

iii) $\sin^2 x = \frac{3}{4}$ $\frac{\sqrt{s}}{r} \left| \frac{A}{c} \right|$

$$\sin x = \pm \frac{\sqrt{3}}{2}$$

Acute $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$ 2

$x = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

c) i) $p = \frac{-360}{2} = 180^\circ$ 1

$a = 3$ 1

